

RESEARCH ARTICLE

Design of a CNN–Swin Transformer Model for Alzheimer’s Disease Prediction Using MRI Images

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ABSTRACT Alzheimer’s Disease (AD) is a progressive neurological condition that deteriorates memory, cognition, and behavior, especially in older adults. Timely identification is essential to enhance patient outcomes and inform therapy choices. This research introduces an extensive deep learning model for multiclass Alzheimer’s disease stage classification using structural MRI images from the publicly accessible OASIS dataset. The procedure begins with thorough preprocessing, including skull stripping, axial slice extraction, and intensity normalization to guarantee uniform input quality. Deep Convolution Generative Adversarial Network (DCGAN) is used to produce authentic synthetic MRI slices, therefore addressing data imbalance and enhancing class representation and training stability. The EffSwin-XNet model introduces a novel hybrid deep learning framework that strategically fuses EfficientNet-B0 and the Swin Transformer, enabling both local and global feature extraction from MRI brain images. This represents a significant advancement over conventional convolutional neural networks and a feature fusion attention mechanism that adaptively emphasizes discriminative features. Grad-CAM is used for explainability to view the brain areas influencing each categorization judgment, hence increasing therapeutic confidence. The model attains a classification accuracy of 95.3%, surpassing traditional CNN and hybrid benchmarks. This study presents an enhanced, interpretable, and efficient method for stage-wise categorization of Alzheimer’s Disease, demonstrating significant potential for use in clinical decision-support systems for early classification and intervention.

INDEX TERMS Alzheimer’s disease, Swin Transformers, explainable AI, deep learning, feature extraction, MRI images.

I. INTRODUCTION

Alzheimer’s disease (AD) is a severe neurodegenerative disorder that impairs brain function and behavior and causes cell destruction [1]. Memory is an important part of the human brain that stores important information that enables us to do simple things like feel emotions and make decisions [2]. AD is a progressive disorder, which implies that patients’ symptoms deteriorate over time. Symptoms take an extended period (around 20 years) to manifest in patients. According to a study published in 2023, over 8.8 million Indians aged 60 and above had dementia as of 2019. The Longitudinal Aging Study in India (LASI) indicated a dementia prevalence of 7.4% among those aged 60 and beyond, with variability

across several states and demographics. India’s elderly population is expected to be 340 million people by 2050. This change in demographics is likely to lead to a big rise in dementia cases, which shows how important it is to improve public health programs and support systems. Treatment costs for AD and other dementias are expected to be over 47 billion dollars [3]. Typically, participants are divided into three groups. The first group is Cognitively Normal (CN), which denotes the lack of cognitive impairment, except for the normal decline associated with aging. The second category is Mild Cognitive Impairment (MCI), a stage between CN and AD. Subjects with CN can progress to MCI when their symptoms become modest but evident. MCI is the earliest clinically apparent development of Alzheimer’s disease. Patients with Mild Cognitive Impairment (MCI) are at a higher risk of developing Alzheimer’s disease

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and receive special attention, as there are currently no disease-modifying treatments available to slow progression and alleviate symptoms [4]. The amount of brain shrinkage seen in the hippocampus and in the temporal lobes gets worse as Alzheimer's disease gets worse. Longitudinal studies that use Magnetic Resonance Imaging (MRI) scans make it easier to find people who are likely to get the condition over time. Neuroimaging techniques have come a long way in recent years, and this has led to the generation of large, multimodal datasets in the field. As a result, more and more neuroscience researchers are making their datasets available to the public. As a result, it is now much easier to get to brain imaging datasets that can help find AD. A lot of research has shown that MRI is a common imaging technology utilized in clinical practice. MRI scans of adult brains can correctly predict AD [5]. We think this because MRI images show that injured cells are not bright [6].

Thayumanasamy and Ramamurthy [7] utilized brain MRI images to evaluate machine learning and deep learning models for detecting Alzheimer's disease. They analyzed designs such as DenseNet-169, ResNet-50, and VGG-16. Their experiments revealed that DenseNet-169 had the highest classification accuracy (82.2%), indicating that it might be a solid starting point for AD detection. However, since the research only used one dataset, the conclusions may be difficult to extend to other groups of individuals. Furthermore, the absence of interpretability mechanisms and transformer approaches for global feature extraction may make it difficult for clinicians to use the suggested model in reality. Salami et al. [8] developed a robust Clinical Decision Support System (CDSS) that leverages deep learning on the massive Open Access Series of Imaging Studies-3 (OASIS-3) neuroimaging dataset to improve Alzheimer's disease diagnostic accuracy and reliability. By integrating pre-processing, feature extraction, and machine learning classification, the technique facilitates early classification of issues and the making of sound therapeutic judgments. The model is mostly based on structural MRI data from previously trained models; however, it lacks a transformer for feature extraction and GradCAM for data interpretation. Pan et al. [9] developed an adaptive ensemble model that combines 3D Convolutional Neural Networks and a Genetic Algorithm (3DCNN+EL+GA) to assist physicians in detecting Alzheimer's disease and moderate cognitive impairment early. This model not only performed better in categorizing data from the ADNI and OASIS datasets, but it also identified key brain regions associated with early AD symptoms, such as the rostral hippocampus and medial amygdala. In contrast, the model extracts features without utilizing a transformer.

Ravi et al. [10] performed a comparison analysis using 26 pre-trained Keras models on ADNI sMRI images for multi-stage Alzheimer's disease classification, revealing that ResNet-50v2 attained the highest accuracy at 91.84%. Grad-CAM and Saliency Maps were used to enhance

interpretability. However, comprehensive in its assessment and stage categorization, it did not utilize fusion-based or GAN-augmented techniques and lacked sophisticated spatial transformers or attention modules, which are integral to enhancing performance. Dardouri [11] suggested a multi-branch CNN capable of using MRI images to detect Alzheimer's disease with an accuracy of 99.68%. The model lacks any Transformer-based components and only uses conventional augmentation techniques, which hinders its ability to learn global properties. Rafsan et al. [12] conducted a comprehensive analysis of deep learning models, including Convolutional Neural Networks (CNN), Bayesian CNN, and U-Net, using the OASIS brain MRI dataset to evaluate their efficacy in classifying Alzheimer's Disease. They used SMOTE-Tomek to rectify the class imbalance and utilized Grad-CAM to enhance model interpretability. The Bayesian CNN had the better accuracy among the investigated models, with 95.03%, indicating its applicability for diagnostics that include ambiguity. However, their work only examined CNN-based designs and did not include more sophisticated and computationally efficient feature extractors such as EfficientNet or contemporary transformer models like the Swin Transformer. The model failed to capture several spatial connections crucial for sophisticated medical image processing due to the absence of attention mechanisms or hybrid architectures.

Hussain et al. [13] proposed a CNN-based approach for identifying Alzheimer's disease using transfer learning with pre-trained models such as AlexNet, GoogleNet, and MobileNetV2. Their methodology achieved exceptional accuracies (up to 99.4% on the Kaggle dataset and 98.2% on the OASIS dataset) and used Grad-CAM to enhance interpretability of the findings. They evaluated the performance of each class and assessed several optimizers. The research only examines CNN-based models and neglects transformer designs or hybrid models, which may enhance spatial attention and feature extraction. Ching et al. [14] developed a GAN-based framework to address the data deficiency in classifying Alzheimer's disease using MRI images. They integrated a pre-trained EfficientNet model with GAN-generated data to execute a multi-stage classification (AD, MCI, NC), achieving enhanced validation accuracy (up to 85.1%). Although they used a GAN-augmented dataset variety, the model architecture was constrained to CNNs, without the incorporation of transformer-based modules or attention mechanisms that may boost spatial awareness and global contextual comprehension. Demir and Selvitopi [15] used conventional machine learning models (e.g., SVM, Random Forest, ANN) to forecast phases of Alzheimer's disease using OASIS and Kaggle datasets. Their research highlighted thorough preprocessing and performance evaluation across models, attaining an accuracy of 85%. However, although the baseline modeling is positive, the work does not use deep learning architectures or classy augmentation techniques or interpretability approaches like Grad-CAM, hence limiting

its applicability to more intricate and high-resolution MRI data.

Although the prior literature has shown a significant amount of work in the classification of Alzheimer's Disease using traditional ML techniques and CNNs, there are still limitations in the interpretability of the model and the global feature modeling. We hypothesize that our hybrid architecture, which integrates EfficientNetB0 with Swin Transformer and is bolstered by DCGAN augmentation and Grad-CAM, will eliminate these deficiencies and surpass extant models in terms of clinical explainability and accuracy on MRI datasets.

The main contributions of this paper are summarized below:

- 1) Proposed a novel deep learning model for predicting Alzheimer's disease from MRI images by merging EfficientNetB0 and Swin transformer with an attention mechanism (EffSwin-XNet).

- 2) Deep Convolutional Generative Adversarial Network (DCGAN) based data augmentation was employed for dynamic hyperparameter tuning and to achieve reliable performance.

- 3) Gradient-weighted Class Activation Mapping(Grad-CAM) visual explanations and adaptive feature-fusion attention at the feature level improve spatial focus and model interpretability.

- 4) A comparative analysis was also conducted using state-of-the-art research in conjunction with the proposed model.

The study's structure is organized as follows: Section II discusses related works on Alzheimer's disease. The Proposed model delineates the workflow, incorporating feature selection, classification, and Grad-CAM interpretability, presented in section III. Experimental results demonstrate that DL outperforms alternative approaches, and the discussion examines findings and model comparisons in Section IV. The Conclusion encapsulates contributions and proposed avenues for future work in Section V.

II. RELATED WORKS

The literature review section examines current methodologies and recent progress in the prediction of AD, emphasizing neuroimaging techniques, including MRI, and the use of deep learning models. This encompasses the examination of diverse architectures, data modalities, and feature extraction techniques used to improve early prediction. Furthermore, we analyze significant hurdles and constraints in contemporary research, including restricted dataset variety and difficulties with model interpretability.

Appiah and Nazir [16] used a multimodal machine learning methodology for the early identification of dementia, amalgamating clinical data with MRI using models such as DenseNet121, ResNet50, and SHAP for interpretative clarity. Their optimal model attained 90% accuracy. Although using multimodal data and explainable AI, the work is deficient in hybrid architectures and Transformer-based models, and fails to integrate GANs or dynamic tuning techniques to mitigate

data scarcity and performance variability. Alahmed and Al-Suhail [17] introduced AlzONet, a deep learning convolutional neural network model trained using several optimizers (Adam, SGD, RMSProp) on the Kaggle MRI dataset. The model attained an accuracy of 96.5% and surpassed well-known pre-trained CNNs in multiclass Alzheimer's disease categorization. Despite AlzONet's strong performance, it only utilizes CNNs, neglecting attention-based or Transformer architectures, and employs only conventional augmentation techniques, hence limiting spatial feature generalization and model interpretability. Jabbar et al. [18] established a CNN-based framework for multiclass Alzheimer's classification using MRI and DenseNet169, attaining a high accuracy of up to 99.05%. They implemented conventional preprocessing and fundamental augmentation to enhance performance. The model does not include sophisticated methodologies like as Swin Transformers or hybrid structures, and it lacks feature-level attention and Grad-CAM representations, therefore limiting both global feature acquisition and interpretability. Venkat et al. [19] presented a novel dual-branch CNN architecture, Alzh-Net, which incorporates multi-scale feature extraction inspired by U-Net, for the automated categorization of Alzheimer's disease using MRI data. It exhibits exceptional accuracy in both binary and multi-class classification tasks, integrating enhanced and multi-view MRI data to improve diagnostic precision. Notwithstanding its superior performance, the model relies on 2D MRI images, which may inadequately represent the brain's intricate structure compared to 3D imaging. Furthermore, the use of augmented datasets may not accurately represent real-world clinical variability, and the model's generalizability across various populations and imaging devices has yet to be evaluated.

Emad et al. [20] proposed a novel meta-learning framework that utilizes Neural Processes (NPs) and transformer encoders to simulate temporal relationships in Alzheimer's disease progression, leveraging the TADPOLE dataset. The model captures individual patient trajectories with greater accuracy, recall, and precision than existing techniques. Class imbalance in the dataset may affect the model's performance and the dependability of its predictions. However, there is a lack of hybrid deep learning models. Jumaili et al. [21] introduced a deep ensemble architecture that integrates VGG16, VGG19, ResNet50, InceptionV3, and EfficientNetB7 for the classification of Alzheimer's disease from MRI images, with a high accuracy of 86.6% using soft voting. The model adeptly used transfer learning and was verified using both internal and external datasets (OASIS and ADNI), demonstrating strong performance. Nonetheless, it did not include sophisticated attention processes or Transformer-based models for refining spatial features. Furthermore, the absence of DCGAN-based augmentation or interpretability methods such as Grad-CAM restricts the model's explainability and flexibility. Jahan et al. [22] introduced a privacy-preserving framework that employs Federated Learning and Explainable AI (SHAP) using multimodal data (MRI, clinical,

psychological) for the identification of Alzheimer's disease, attaining better accuracy. The Random Forest-based federated learning approach enhances interpretability and safeguards data privacy across institutions. Nevertheless, it is devoid of deep learning architectures such as CNNs or Transformers and fails to use spatial attention for MRI image processing, hence constraining spatial feature learning and model scalability in visual data. Vanaja et al. [23] created a tailored Deep Convolutional Neural Network (cDCNN) using SMOTE for the classification of Alzheimer's disease from MRI datasets (ADNI and Kaggle), with an accuracy of 87% despite preprocessing difficulties. Their comparative investigation using machine learning techniques underscores the efficacy of convolutional neural networks in the early diagnosis of Alzheimer's disease. The architecture does not use attention mechanisms or hybrid deep models, and it does not utilize interpretability approaches such as Grad-CAM, which restricts transparency.

Sathyabhama et al. [24] introduced an Exponential Gaussian Error Linear Unit–Squeeze Net (EGELU-SZN) classifier for the early diagnosis of Alzheimer's disease, using skull stripping Adaptive Median Otsu's Thresholding (AMOT), segmentation Rectilinear Mayfly Optimization-centric Automatic Seeded Region Growing Algorithm (RMF-ASRG), and feature reduction Reflective Correlation Principal Component Analysis (RCPKA) techniques on MRI images. The model achieved impressive performance (accuracy: 95.99%) on ADNI data via an innovative deep learning approach. Nevertheless, it is devoid of attention processes akin to Transformers and interpretability tools such as Grad-CAM, hence constraining explainability. Awang et al. [25] performed a thorough assessment of deep learning techniques used in 3D imaging for Alzheimer's classification, emphasizing the prevalence of CNN and 3D-CNN architectures, as well as the promise of multimodal integration and attention processes. The paper highlights the benefits of volumetric imaging and contemporary developments in transfer learning and Generative Adversarial Networks (GANs). Nonetheless, it neither proposes nor assesses a cohesive hybrid model, nor does it provide empirical validation for sophisticated designs such as Transformers or hybrid fusion models. Al-Shoukry et al. [26] suggested a Hybrid AI model that integrates machine learning (SVM, CatBoost, XGBoost) and deep learning (Stacked CNN) for the early identification of Alzheimer's disease using MRI, attaining an accuracy of 94.22%. SMOTE and data augmentation improved equilibrium and resilience. Nonetheless, Transformer-based models were not used, and interpretability via attention mechanisms or Grad-CAM was not investigated, hence limiting explainability. Rajasree and Brindha [27] formulated a four-phase ensemble model integrating Bi-GRU, MLP, and QDNN, optimized by EMOAOA, for the early diagnosis of Alzheimer's disease utilizing data derived from statistical and correlation-based methodologies. The model attained better accuracy via proficient feature ranking and optimization

of ensemble weights. However, it neither employs imaging data directly nor integrates spatial attention or transformer-based designs, hence limiting its use to MRI-based spatial diagnostics.

Castellano et al. [28] devised a multi-modal Alzheimer's disease prediction system that combines 3D MRI with amyloid Positron Emission Tomography (PET), using convolutional neural networks and Grad-CAM to enhance accuracy and interpretability, attaining state-of-the-art performance on OASIS-3. The methodology prioritizes volumetric imaging and interpretability. However, it does not include hybrid fusion models to improve performance. Hu et al. [29] introduced Conv-Swinformer. This hybrid deep learning model integrates Convolutional Neural Networks (CNN) with a shift window attention mechanism to enhance local feature extraction in 3D MRI data for Alzheimer's disease classification. This architecture effectively captures fine-grained lesion features by focusing attention on small spatial regions, resulting in improved classification performance compared to traditional models. However, while the model demonstrates better performance, further validation is necessary to assess its interpretability. Fathi et al. [30] employed conventional architecture and transfer learning. The paper presents the CNN-based ensemble method (WPBEM) for classifying Alzheimer's disease; however, it does not utilize contemporary transformer-based models or explainability mechanisms. Illakiya and Karthik [31] model incorporates Swin Transformer, DCPAN, and an Age Deviation Factor for pMCI and sMCI classification; it is devoid of DCGAN-based augmentation and Grad-CAM visual explainability, all of which are essential for clinical trust. Kang et al. [32] proposed a 3D GAN-based three-round learning framework for AD staging with Grad-CAM and transfer learning, which is presented in this study; however, it lacks architectural variety and only uses CNN-based feature extraction. Additionally, it lacks hybrid backbones like Swin Transformer for global context modeling and sophisticated attention mechanisms. Illakiya and Karthik [33] study primarily focuses on combining global context and cross-dimensional features for MCI classification; however, hybrid transformer-CNN architectures and DCGAN-based augmentation, which can help address training instability and data imbalance, are not integrated. Additionally, explainability techniques like Grad-CAM and dynamic attention fusion are not included. Wang et al. [34] utilized cross-dimensional feature fusion and global context learning for MCI classification but did not incorporate hybrid CNN-Transformer architectures, sophisticated attention mechanisms, or DCGAN-based augmentation to address class imbalance. It also lacks the visual explainability of Grad-CAM, which is critical for clinician trust.

Although deep learning models have made progress in the classification of Alzheimer's disease, there are still substantial deficiencies, particularly in the accurate collection of both local and global brain features, the

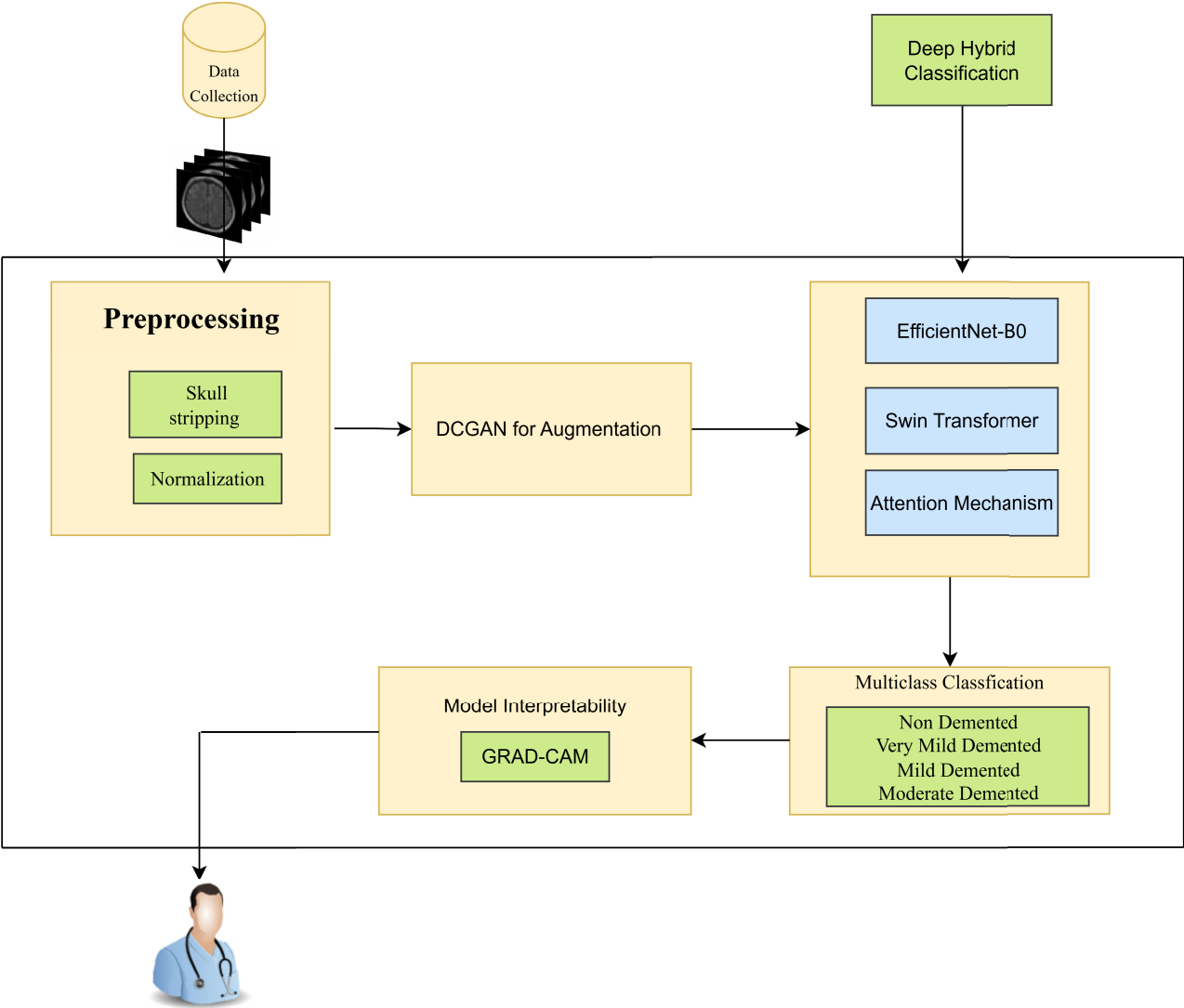


FIGURE 1. Pictorial representation of the proposed model.

management of small datasets, and the assurance of model interpretability, according to the research. Numerous studies neglect region-specific information that is essential for early diagnosis. The proposed model combines Swin Transformer for global contextual feature acquisition with EfficientNetB0 for local feature extraction. It is further enhanced by attention mechanisms, DCGAN-based augmentation, and Grad-CAM for interpretability and robustness.

III. PROPOSED EffSwin-XNet: A MODEL COMBINING EfficientNetB0, SWIN TRANSFORMER, ATTENTION, AND INTERPRETABILITY

In this part, Figure 1 the process and materials utilized to multiclass categorize OASIS MRI data for AD are detailed. An attention-based fusion mechanism, EfficientNet-B0, Swin Transformer, and data augmentation utilizing

DCGenerative Adversarial Networks (DCGANs) are all part of the proposed model innovative hybrid deep learning model. Thorough preprocessing is also done. In contrast to earlier research on CNN-based models that focused on performance like VGGNet, ResNet, and DenseNet, this study introduces EfficientNet-B0, a lightweight design known for its computational efficiency in Alzheimer’s disease categorization. Using EfficientNet-B0 as a foundation for low-level feature extraction, the model is able to detect significant local abnormalities in MRI scans of the brain that are associated with alterations in the structure of the ventricles, entorhinal cortex, and the hippocampus. Furthermore, the Swin Transformer, a hierarchical vision transformer, captures global context and long-range interconnectedness, which are commonly overlooked in traditional CNNs. Integrating these two architectures creates a multi-scale feature extraction

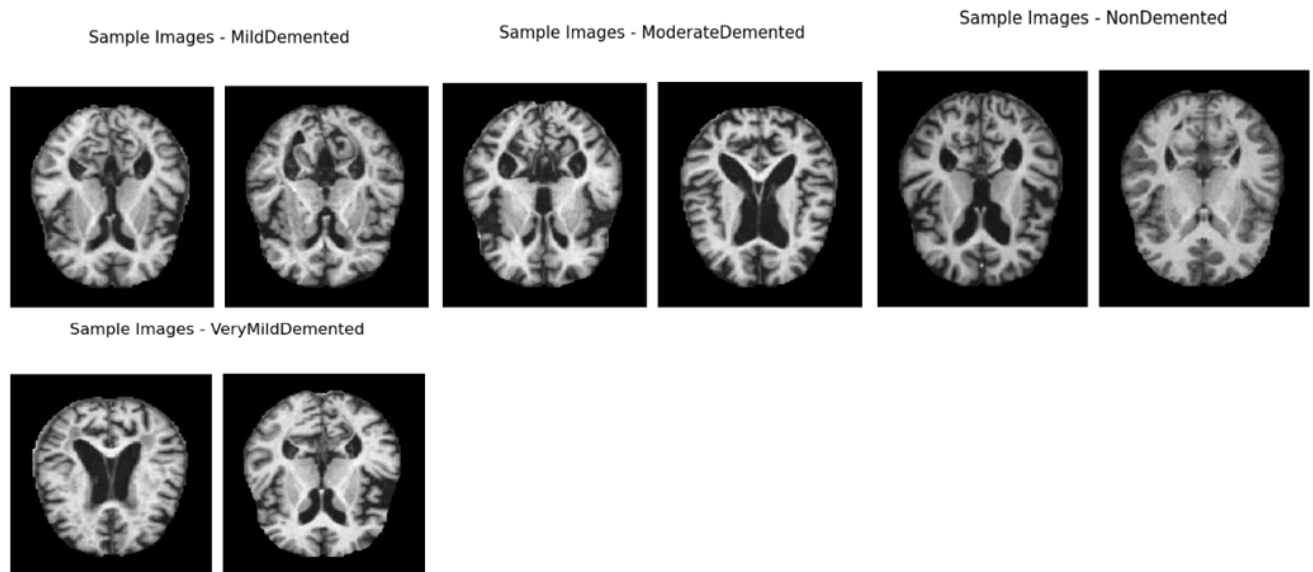


FIGURE 2. Sample MRI slices from each class of the Alzheimer's dataset: MildDemented, ModerateDemented, NonDemented, and VeryMildDemented (2 images per class).

TABLE 1. Demographic and clinical characteristics by CDR score.

CDR Score	Dementia Status	Subjects	Mean Age (years)	Sex (M/F)	Education (years)	MMSE Score (Mean $\pm$ SD)
0	Non-Demented	325	44.38 $\pm$ 24.15	120 / 205	3.45 $\pm$ 1.22	28.97 $\pm$ 1.21
1	Very Mild Demented	82	71.03 $\pm$ 16.20	38 / 44	2.89 $\pm$ 1.27	26.00 $\pm$ 3.10
2	Mild Demented	27	76.55 $\pm$ 12.69	18 / 9	2.48 $\pm$ 1.34	23.00 $\pm$ 3.30
3	Moderate Demented	2	82.00 $\pm$ 5.65	1 / 1	2.00 $\pm$ 1.41	15.00 $\pm$ 0.00

approach suitable for detailed local features and large structural patterns in the brain. Extracted features are fused and then given to the attention mechanism technique to extract essential features for classification. Also included Gradient-weighted Class Activation Mapping (Grad-CAM) to enhance model interpretability. The pipeline is designed to categorize individuals into four primary categories: Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented.

Dataset Description: Researchers used a lot of different datasets to learn more about AD. For this study, we used the OASIS dataset. The Open Access Series of Imaging Studies (OASIS) dataset [35] has neuroimaging datasets that are publicly available for study and development. Researchers can get a full image of Magnetic Resonance Imaging (MRI) data from healthy people and people with different neurological disorders through the OASIS project.

Figure 2 shows the MRI slices demonstrate modest structural variations between stages of dementia. Non-demented brains have brighter and more intact architecture, while moderately demented samples reveal significant atrophy and darker areas. Mild and very mild demented images exhibit intermediate patterns, reflecting gradual alterations in brain structure associated with disease severity. The dataset contains MRI brain scans, demographic information, and clinical assessments from both healthy persons and patients

who have been diagnosed with Alzheimer's disease. Most of it is made up of structural MRI scans, which frequently contain T1-weighted images. We can learn a lot about how the brain is built together from these images.

Table 1 shows that the dataset for Alzheimer's patients may also include clinical evaluations of cognitive function, disease progression, or other relevant metrics.

Table 1 delineates We used the OASIS dataset, which is a publicly available collection of cross-sectional T1-weighted MRI scans, and only looked at people 60 and older to make sure it was relevant to Alzheimer's disease. Each person in the dataset has a Clinical Dementia Rating (CDR), which makes it easy to put them into one of four groups:

- Non-Demented (CDR = 0),
- Very Mild Demented (CDR = 1),
- Mild Demented (CDR = 2), and
- Moderate Demented (CDR = 3).

This enhanced categorization facilitates the identification of nuanced cognitive alterations, augmenting the model's capacity for early and precise diagnosis.

A. DATA PREPROCESSING

Classification of Alzheimer's disease using Magnetic Resonance Imaging (MRI) requires a thorough preprocessing workflow to ensure consistent feature extraction across

subjects. As part of this process, scale the images, normalize their intensity, remove slices from the skull, and extract slices. Prior to isolating vital brain regions like the hippocampus and lateral ventricles essential for dementia diagnosis, the Brain Extraction Tool (BET) is used to remove non-brain components, such as the skull, scalp, and cerebrospinal fluid. After that, 2D axial slices are taken from each 3D T1-weighted MRI scan, with a focus on the middle 30–40 slices that show the medial temporal lobe, a first site of Alzheimer’s disease. The slices are consistently scaled to 224×224 pixels to standardize model inputs. To mitigate intensity variability arising from disparate scanning protocols, a two-step normalization process is employed: z-score standardization (subtracting the mean and dividing by the standard deviation of voxel intensities), followed by min-max normalization to scale the data within the $[0, 1]$ range, thereby improving training stability. The dataset is divided into 80% for training and 20% for testing, using subject-wise stratified sampling to ensure class equilibrium and avert data leakage. All scans from a single person are restricted to a single subgroup, and the entire procedure is replicated across five random seeds to ensure robustness and present averaged performance metrics. Elimination of noise and slice selection are essential for magnetic resonance imaging. Magnetic resonance images often exhibit intensity fluctuations due to variations in Radio Frequency coil sensitivity, magnetic field inhomogeneity, and equipment-related issues. To address this problem in the subsequent phase, performed bias field correction using a non-parametric, non-uniform intensity normalization technique from the Advanced Normalization Tools (ANTs). Subsequently, enumerated the slices in each research individually. Additionally, we determined the mode of each study and selected median slices, along with slices representing two standard deviations above and below the median.

B. DATA AUGMENTATION-DEEP CONVOLUTIONAL GENERATIVE ADVERSARIAL NETWORK (DCGAN)

To mitigate the problem of constrained and uneven data, a Deep Convolutional Generative Adversarial Network (DCGAN) was used. The DCGAN is trained to produce realistic 2D MRI slices based on class labels, resulting in semantically coherent and physically credible visuals for underrepresented categories. The illustration representing class balance before and after augmentation using DCGANs is a crucial component of the proposed deep hybrid model for Alzheimer’s disease classification.

Figure 4 shows sample MRI slices generated by a Deep Convolutional GAN (DCGAN) trained on the Alzheimer’s dataset. These synthetic images closely replicate the structural and textural patterns of real scans, helping to address data scarcity and class imbalance by enriching the training set without additional manual labeling.

Figure 5 shows the initial class distribution in the training dataset showed a significant imbalance, with the “Non-Demented” category having 2,600 images, while the

“Mild Demented” and “Very Mild Demented” categories had 727 and 1,792 images respectively. The “Moderate Demented” category was significantly underrepresented, with only 52 samples. To address this, a DCGAN-based augmentation technique was used to achieve a consistent class size of 2,600 samples for each class, as shown in Figure 5. The resulting dataset now displays a fully balanced class distribution, mitigating class bias and improving generalization performance across all categories. This transformation ensures equal contribution from each class during training, allowing the model to acquire representative features from all diagnostic categories.

DCGAN is composed of two fundamental components: a Generator that employs transposed convolutional layers to convert random noise (latent vectors) into synthetic images, and a Discriminator that employs convolutional feature extraction to distinguish between real and generated images. DCGANs provide structurally coherent images with more stable training dynamics, in contrast to traditional GANs that frequently generate indistinct or unstable outputs due to their dependence on dense layers, as shown in Figure 3.

DCGAN is a potent data augmentation tool that effectively addresses data scarcity and class imbalance in the context of AD stage classification using MRI images. To learn stage-specific anatomical patterns (Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented), the augmentation procedure commences with the training of a DCGAN on real 2D MRI segments (e.g., axial T1-weighted images) from the OASIS dataset. This is done class-wise. The generator generates synthetic MRI slices that accurately represent the structural characteristics of each AD stage after it has been trained. These synthetic samples are subsequently combined with the original dataset, with a particular emphasis on enriching underrepresented classifications such as Moderate Demented, which may have as few as 52 actual samples. DCGAN guarantees a balanced dataset for training by producing thousands of new images (e.g., increasing the number of samples in each class to 2,600).

The EffSwin-XNet model, which is a hybrid of EfficientNetB0, Swin Transformer, and Attention mechanisms, is subsequently trained using the augmented dataset. This model is capable of learning from both real and synthetic data. The following are several critical advantages of this augmentation strategy: it preserves biologically meaningful structures, such as hippocampal atrophy or cortical thinning, avoids unrealistic distortions associated with basic augmentation methods like flipping or rotating, and improves the model’s capacity to generalize to unseen MRI patterns. DCGAN is a fundamental component of AD classification pipelines that are both interpretable and robust, as it effectively mitigates class imbalance, enriches limited datasets, reduces overfitting, and enhances diagnostic performance. Figure 3 shows the DCGAN architecture used in this work, where the generator synthesizes realistic MRI images through transposed convolutions and residual

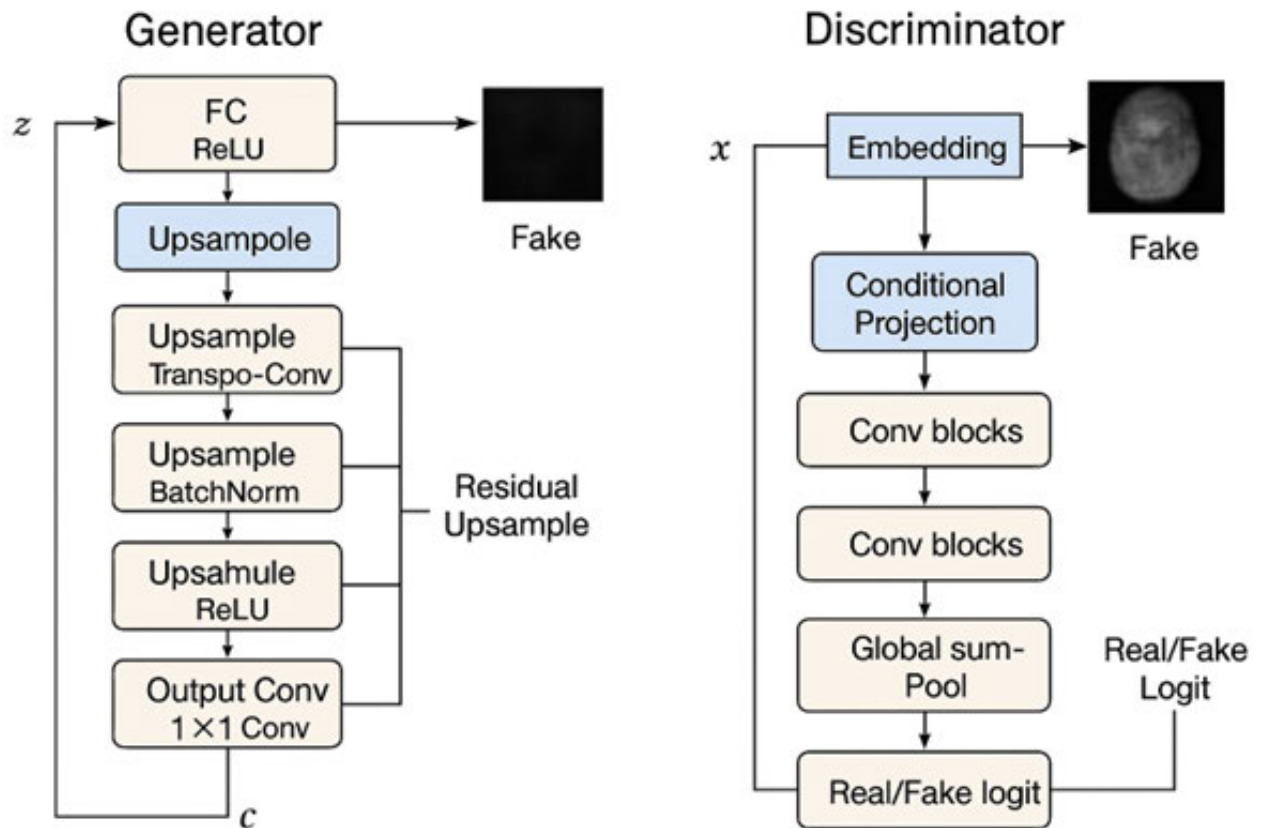


FIGURE 3. Deep convolution generative adversarial network architecture.

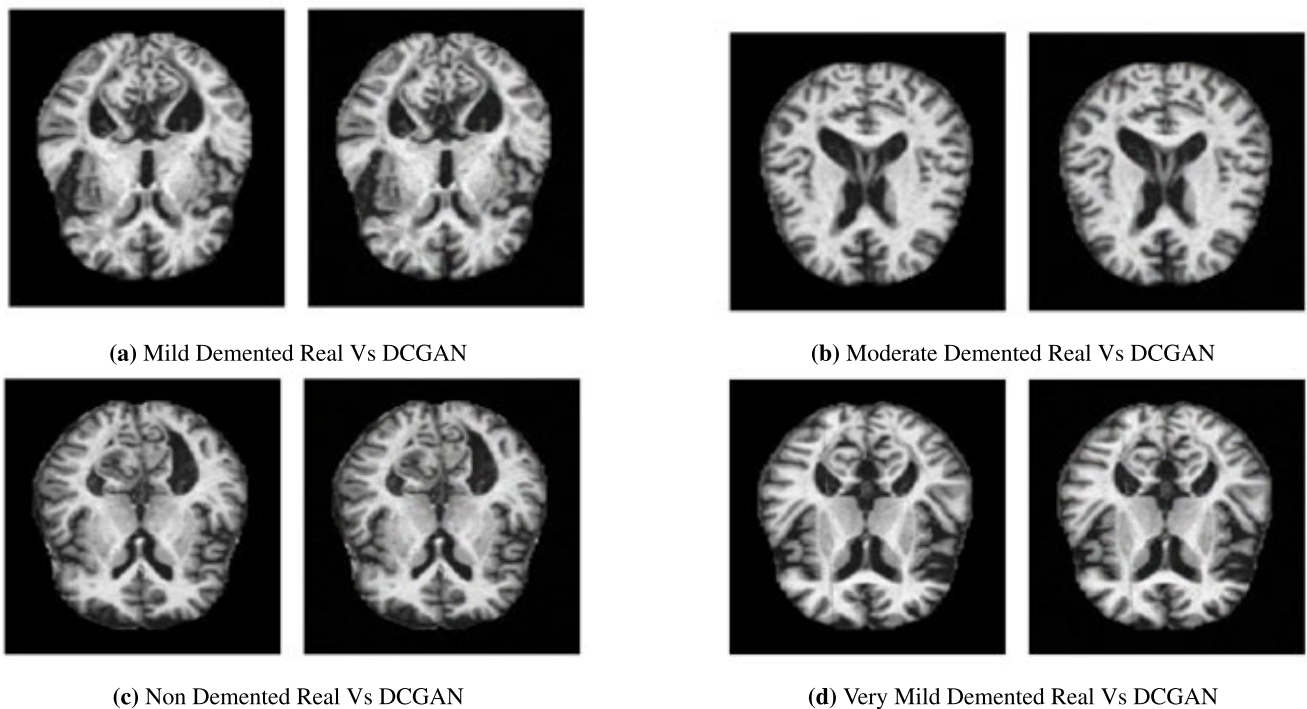
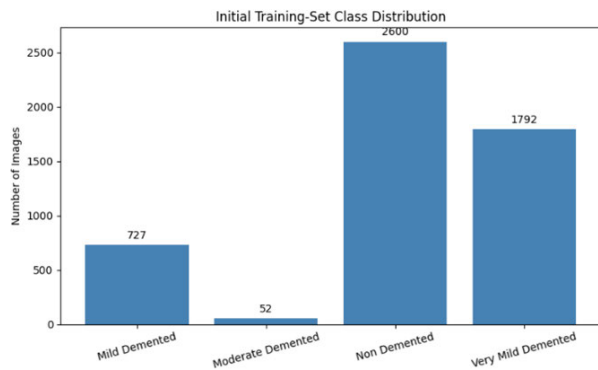


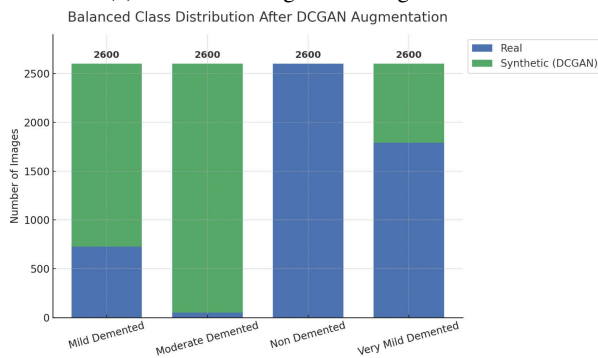
FIGURE 4. Sample results of real and DCGAN-generated MRI slices for different stages of Alzheimer's Disease.

upsampling, while the discriminator employs conditional projections and convolutional layers to distinguish between

real and synthetic slices, aiding in class balancing and enhancing model robustness (Goodfellow et al., 2014;



(a) Class Balancing before Augmentation



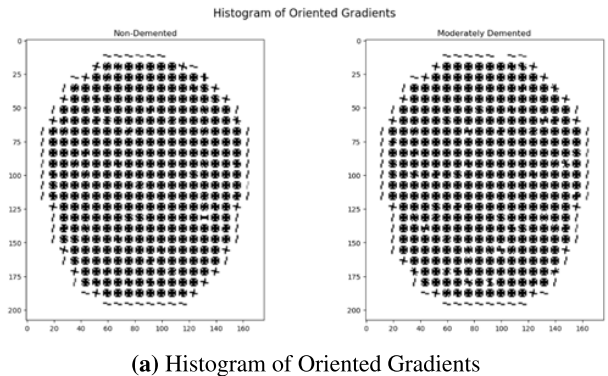
(b) Class Balancing after Augmentation

FIGURE 5. Class balancing before and after augmentation.

Radford et al., 2016). Compared to conventional GANs, DCGANs are particularly well-suited for image generation due to their use of convolutional layers, which results in superior spatial feature preservation and structurally consistent MRI synthesis.

C. HISTOGRAM OF ORIENTED GRADIENTS (HOG) AND PRINCIPAL COMPONENT ANALYSIS (PCA)

The Figure 6 shows a comparison of the Histogram of Oriented Gradients (HOG) from brain MRI scans of two groups of people: those who are not demented and those who are somewhat demented. HOG is a strong feature descriptor that captures the structure and texture of an image by documenting how gradient orientations are spread out in different places. In this Figure 6, each arrow shows the main direction of the change in intensity (gradient) in each cell of the image. By looking at the direction and strength of these gradients in space, you may see subtle changes in the patterns of brain tissue that may be linked to different stages of dementia. The picture shows that this kind of difference map shows parts of the brain that alter shape as dementia becomes worse. Big variations in areas like the ventricles, brain, and white matter might indicate that Alzheimer's Disease is causing tissue to break down or shrink. These insights are important for figuring out how neurodegeneration works and may make it easier to create features for classification.



(a) Histogram of Oriented Gradients

Difference Between NonD & ModerateDemented Mean



(b) Contrast difference between Non-Demented and Moderate Demented Mean

FIGURE 6. Histogram and contrast.

The Figure 6 shows MRI scans in detail that are meant to tell the difference between those who are not demented and those who are somewhat impaired. The top row shows Histogram of Oriented Gradients (HOG) visualizations for brains that are not demented and those that are slightly demented. These visualizations gather edge and contour information to show the structural differences between the two groups. Next to this is a difference map that shows how the intensity of pixels varies between the average MRI scans of people with mild to moderate dementia and those without dementia. This map makes it easy to see where neurodegeneration has happened. The lower two rows display eigenimages generated from the principal components of non-demented individuals, illustrating the predominant patterns of anatomical variation obtained using Principal Component Analysis (PCA). These eigenimages serve as a reference for detecting anomalies associated with cognitive deterioration. Collectively, these visualizations provide significant insights into disease-specific structural alterations and offer interpretable elements that may inform and improve deep learning models for the categorization of Alzheimer's disease.

Figure 1 illustrates the proposed pipeline, which demonstrates an efficient and thorough methodology for multiclass categorization of Alzheimer's disease using MRI data. The system tackles issues of data scarcity and intricate feature extraction via the integration of sophisticated preprocessing approaches, DCGAN-based data augmentation, and a robust deep hybrid model that combines EfficientNet-B0, Swin Transformer, and attention mechanisms. Furthermore, the use of Grad-CAM for model interpretability augments trust and transparency, making the forecasts therapeutically significant for end users. This paradigm has the potential to enhance the early and accurate identification of Alzheimer's Disease and associated disorders, facilitating prompt intervention and improved patient outcomes.

D. PROPOSED EffSwin-XNet HYBRID MODEL

The proposed hybrid model strategically combines EfficientNet-B0 with the Swin Transformer to use the advantages of both convolutional and transformer-based approaches for medical image categorization. This complementary integration is further augmented by a specialized attention mechanism that adaptively consolidates diverse information to boost model interpretability and discriminative influence. Also, provided an innovative hybrid architecture that combines the local pattern learning proficiency of EfficientNet-B0 with the long-range dependency modeling of a Swin Transformer, interconnected by an attention-based fusion module for adaptive integration. Figure 7 shows that the EffSwin-XNet architecture takes the best parts of EfficientNet-B0 and the Swin Transformer and uses them to use both local and global feature representations from MRI images. This makes it possible to accurately classify Alzheimer's disease into many classes, as shown in Figure 7. The Swin Transformer gets hierarchical global context via changing windows and multi-stage attention processes, whereas EfficientNet-B0 gets detailed local information through its well arranged convolutional blocks. A feature fusion module with an attention mechanism is used to integrate the complementary features and show the most important information. The combined characteristics are then used for robust multiclass classification, and Grad-CAM makes it easy to understand so that clinical end-users can predict the underlying process. This hybrid technique adeptly reconciles local specificity with global context, enabling better diagnostic accuracy and reliability in predicting Alzheimer's Disease.

1) EfficientNet-B0 AS A LOCAL FEATURE EXTRACTOR

EfficientNet-B0 is the backbone Convolutional Neural Network (CNN) that pulls out low-level and mid-level local features from MRI images. Unlike regular CNNs that scale without regard to the task at hand, EfficientNet-B0 uses a compound scaling strategy that always changes the network's breadth, depth, and input resolution using a set of carefully chosen coefficients. This strategy makes it easier to get better results with a lot fewer factors.

In the realm of MRI-based Alzheimer's Disease diagnosis, EfficientNet-B0 proficiently extracts local spatial features, including tissue intensities, structural boundaries, and localized abnormalities. The design utilizes MBConv blocks, which consist of depthwise separable convolutions enhanced with squeeze-and-excitation modules, allowing for efficient modeling of channel-wise feature dependencies. This design is particularly adept at detecting subtle structural changes in brain areas that often indicate various stages of Alzheimer's Disease. EfficientNet-B0 is used as the primary feature extractor because of its lightweight architecture and superior parameter efficiency. It collects detailed anatomical data from 2D MRI slices while minimizing the processing burden [36].

EfficientNet-B0 employs compound scaling to uniformly scale network dimensions: depth (d), width (w), and input resolution (r). The scaling is defined as:

$$d = \alpha^\phi \quad (1)$$

$$w = \beta^\phi \quad (2)$$

$$r = \gamma^\phi \quad (3)$$

subject to the constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2; \quad \alpha, \beta, \gamma > 0 \quad (4)$$

The primary building block of EfficientNet-B0 is the MBConv layer, which integrates a Squeeze-and-Excitation (SE) block for channel-wise attention. Given an input feature map $X \in \mathbb{R}^{H \times W \times C}$, the SE block computes:

$$S = \sigma(W_2 \cdot \delta(W_1 \cdot \text{GAP}(X))) \quad (5)$$

Here, $\text{GAP}(X) \in \mathbb{R}^C$ denotes the global average pooled vector, $\delta(\cdot)$ is the ReLU activation function, $\sigma(\cdot)$ is the sigmoid function, and W_1, W_2 are learnable weights.

The recalibrated feature map is obtained via channel-wise multiplication:

$$X_{\text{SE}} = X \otimes S \quad (6)$$

The final local output of the EfficientNet-B0 model for an input image I is represented as:

$$F_{\text{local}} = \text{EffNetB0}(I) \quad (7)$$

2) SWIN TRANSFORMER FOR GLOBAL CONTEXTUAL FEATURE EXTRACTION

Swin Transformer to improve the local feature representations from EfficientNet-B0 used. The Swin Transformer is great at modeling long-range connections and global structural patterns in image data. In contrast to the Vision Transformer (ViT), which analyzes entire images in flat patches and exhibits quadratic complexity, the Swin Transformer employs a hierarchical approach by using non-overlapping local windows for image processing. The windows are positioned between layers to facilitate cross-window information exchange, thereby enhancing efficiency and scalability while preserving global awareness. Every level in the Swin Transformer hierarchy expands the receptive area, enabling

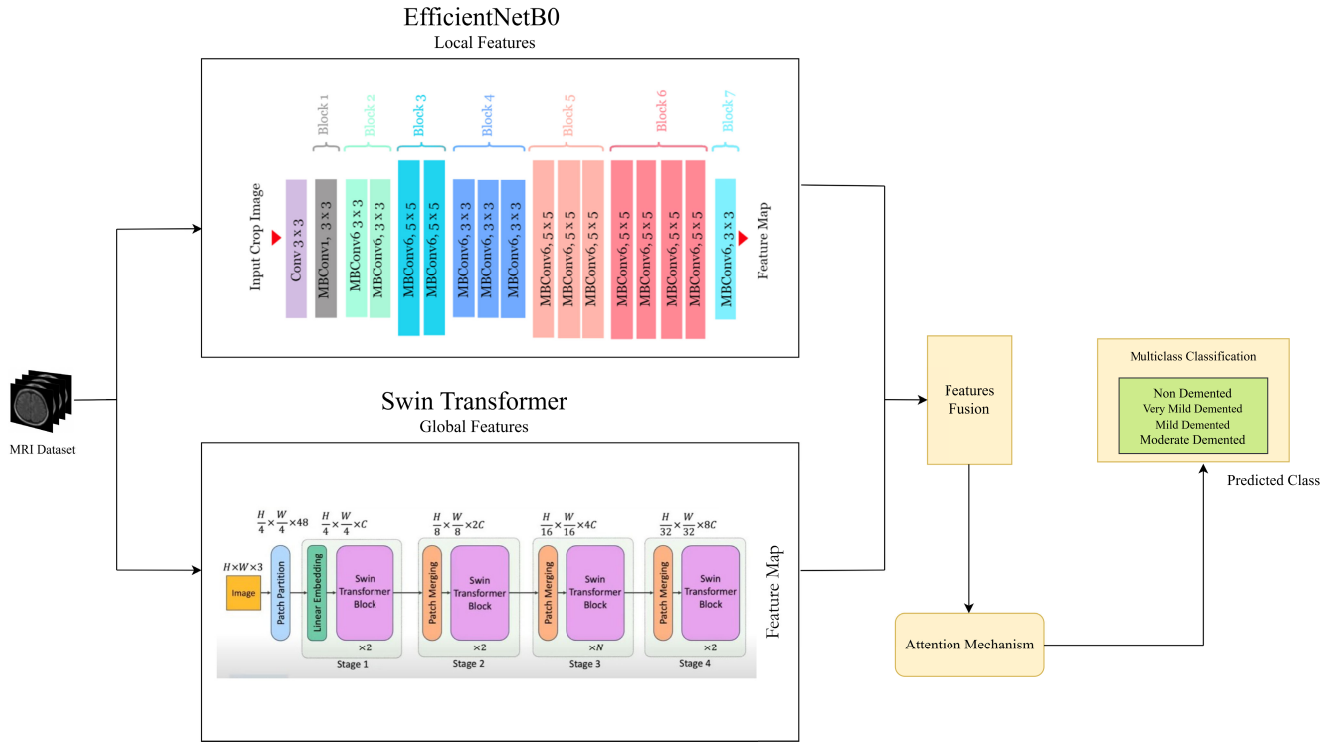


FIGURE 7. Architectural diagram of EffSwin-XNet model.

the model to consolidate input from broader global features. This feature is essential for detecting scattered pathological alterations, like ventricular enlargement, cortical thinning, or global atrophy, key indicators of Alzheimer's Disease development. Include a Swin Transformer module downstream of the CNN to capture global spatial relationships and hierarchical context. The Swin Transformer utilizes shifting window self-attention to represent interactions across spatial patches, thereby contextualizing localized data derived from EfficientNet [37]. The Swin Transformer operates on image patches $P \in \mathbb{R}^{M \times M \times C}$, which are first projected into patch embeddings $z_0 \in \mathbb{R}^{N \times D}$, where N is the number of patches and D is the embedding dimension.

Within each non-overlapping window, self-attention is computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \quad (8)$$

Here, the query, key, and value matrices are derived by linear projections:

$$Q = XW_Q \quad (9)$$

$$K = XW_K \quad (10)$$

$$V = XW_V \quad (11)$$

where X is the input feature matrix, W_Q , W_K , and W_V are learnable weight matrices, and d_k is the dimensionality of the key vectors. The term B denotes the relative positional bias.

To enable cross-window communication, window partitions are shifted between consecutive layers.

The final global feature representation produced by the Swin Transformer for an input image I is given by:

$$F_{\text{global}} = \text{SwinTransformer}(I) \quad (12)$$

3) ATTENTION MECHANISM FOR DYNAMIC FEATURE INTEGRATION

Here, constructed a novel attention mechanism that serves as a fusion layer to integrate the spatially localized features from EfficientNet-B0 with the globally contextualized representations from the Swin Transformer. An attention method integrates outputs from the CNN and Transformer branches. The module calculates attention weights that highlight complementary features and diminish irrelevant activations. This enables the model to focus preferentially on areas of interest, such as the hippocampus and parietal cortex, thereby improving classification performance. The primary objective of this attention module is to dynamically evaluate the significance of each feature stream (local and global) in the classification task. This can be achieved by combining channel attention and spatial attention, as demonstrated by Convolutional Block Attention Modules (CBAM), or through cross-attention mechanisms that explicitly learn the interdependencies between the two feature modalities. The attention process produces a refined, high-level representation by highlighting significant aspects and diminishing duplicate or unnecessary information. This enables the model to focus

on key diagnostic indicators that may be overlooked when using either convolutional or transformer-based models alone [38]. To integrate local and global feature representations, an adaptive attention mechanism is applied. First, the feature vectors are concatenated:

$$\mathbf{F}_{\text{concat}} = [\mathbf{F}_{\text{local}} \parallel \mathbf{F}_{\text{global}}] \quad (13)$$

An attention gate is then computed using a sigmoid-activated linear transformation:

$$\mathbf{a} = \sigma(\mathbf{W}_a \cdot \mathbf{F}_{\text{concat}} + \mathbf{b}_a) \quad (14)$$

Finally, the features are fused in a gated manner:

$$\mathbf{F}_{\text{fused}} = \mathbf{a} \otimes \mathbf{F}_{\text{local}} + (1 - \mathbf{a}) \otimes \mathbf{F}_{\text{global}} \quad (15)$$

Here, $\otimes$ denotes element-wise multiplication, $\mathbf{W}_a$ and $\mathbf{b}_a$ are learnable weights, and $\sigma(\cdot)$ represents the sigmoid activation function.

4) CLASSIFICATION HEAD

The integrated features are transmitted via a fully connected layer, followed by a softmax classifier to ascertain the probability distribution among the categories. The fused vector is passed through a dense layer and a softmax for classification over K AD stages:

The fused feature vector is passed through a classification head consisting of a linear transformation followed by a softmax activation to produce the output prediction:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_c \cdot \mathbf{F}_{\text{fused}} + \mathbf{b}_c) \quad (16)$$

Here, $\mathbf{W}_c$ and $\mathbf{b}_c$ are the learnable weights and bias of the classification layer, and $\hat{\mathbf{y}} \in \mathbb{R}^K$ is the predicted class probability vector over K classes.

The proposed hybrid model integrates EfficientNet-B0 with Swin Transformer via an attention-based fusion technique, successfully capturing both fine-grained local features and holistic global patterns, thereby enhancing performance in the multiclass classification of Alzheimer's disease stages, as shown in Algorithm 1.

E. MODEL TRAINING AND EVALUATION

In deep learning, model training and evaluation are very important for building and improving models that can perform tasks effectively. The Adam optimizer is used for training, and the learning rate starts at $1e-4$ and subsequently goes down according to a cosine annealing schedule. The model is trained for up to 50 epochs, with early termination depending on the validation loss. The loss function is categorical cross-entropy, and class weights are used to help balance out the classes. The dataset has been split into two parts: one for training and one for testing. The training procedure involved refining the model using the cross-entropy loss function and the Adam optimizer over multiple epochs. Cross-entropy loss is used as a statistic to train the model and see how well it worked. During training, the Adam optimizer changes the model's weights. The training loop is

Algorithm 1 Internal Workings of the EffSwin-XNet Model

-
- 0: **Input:** Preprocessed and Augmented MRI Images

 - 0: **Feature Extraction:**
 - 0: **EfficientNet-B0 Branch:**
 - 0: Extract local features using inverted residual blocks and depthwise separable convolutions
 - 0: Output feature map $\mathbf{F}_{\text{CNN}}$
 - 0: **Swin Transformer Branch:**
 - 0: Divide image into non-overlapping patches
 - 0: Process patches via hierarchical Transformer blocks with shifted window attention
 - 0: Capture global contextual features
 - 0: Output feature map $\mathbf{F}_{\text{Swin}}$

 - 0: **Feature Fusion:**
 - 0: Fuse $\mathbf{F}_{\text{CNN}}$ and $\mathbf{F}_{\text{Swin}}$ via concatenation or summation to form $\mathbf{F}_{\text{fused}}$
 - 0: Optionally apply a linear projection to align dimensions

 - 0: **Attention Mechanism (Adaptive Feature Fusion by concatenation):**
 - 0: **Fusion:** Combine local and global features, then fused features are transformed with a sigmoid activation to compute attention weights on $\mathbf{F}_{\text{fused}}$
 - 0: Use these attention weights to adaptively fuse features, emphasizing relevant information.

 - 0: **Classification Head:**
 - 0: Apply Global Average Pooling to $\mathbf{F}_{\text{attention}}$
 - 0: Pass through one or more Dense layers
 - 0: Apply Softmax activation to obtain class probabilities

 - 0: **Interpretability with Grad-CAM:**
 - 0: Generate Grad-CAM heatmaps using gradients from the final convolutional layers
 - 0: Overlay heatmaps on input MRI slices for visualization

 - 0: **Output:**
 - 0: Predicted class label: {Non-Demented, Very Mild Demented, Mild Demented, Moderate Demented}
 - 0: Corresponding Grad-CAM heatmap = 0
-

meant to go over the training dataset for a defined number of epochs, figuring out the loss and accuracy for each one. During training, we employed early stopping with validation loss as the monitoring metric to prevent overfitting. The model's performance on the validation set was monitored at each epoch, and training was stopped if no improvement was observed over a defined number of epochs (patience = 5). Importantly, the model checkpoint corresponding to the lowest validation loss was automatically saved and used for

TABLE 2. Model configuration summary.

Category	Parameter	Value / Setting
2*Architecture	Backbone Models Fusion Layer	EfficientNetB0 + SwinTransformer Fully Connected (512 units, ReLU) + Dropout (p = 0.4)
6*Training	Optimizer Initial Learning Rate Scheduler Weight Decay Batch Size Epochs	Adam 1e-4 Cosine Annealing (T_max = 50) 1e-4 32 20 Epochs
2*Loss Function	Type Class Balancing	Weighted Cross-Entropy Inverse frequency-based class weights
2*Data	Input Size Augmentations	224 × 224 RGB Crop, Flip, Distortion, Noise, Brightness, Normalize
Validation	Cross-Validation	5-fold stratified cross-validation
Explainability	CAM Method	Grad-CAM applied to the final convolutional layer
Evaluation	Metrics	Accuracy, Precision, Recall, F1-score

the final evaluation on the test set. This approach ensured that the reported results reflect the best generalization capability of the model rather than relying on the final training epoch. A confusion matrix has been made to show how well the classification works, and the accuracy, precision, recall, and F1-score have all been calculated. To guarantee the robustness of the model evaluation and to prevent data leakage, the dataset was divided into a training-validation set and a dedicated testing set. Specifically, 80% of the data was utilized for internal validation and training, while the remaining 20% was set aside as an independent testing set for the final performance assessment. In order to optimize model parameters and reduce overfitting, we implemented 5-fold stratified cross-validation on the 80% of the training phase. The testing set was exclusively reserved for the reporting of final classification accuracy, precision, recall, F1-score, and Grad-CAM-based interpretability, and was not utilized during any stage of training. This approach ensures that performance measurements are impartial and consistent with the most effective methodologies for clinically focused machine learning applications.

Table 2 illustrates the meticulous adjustment of critical hyperparameters, including a modest learning rate with cosine annealing, class-balanced cross-entropy loss, and dropout regularization, which facilitated resilient training and generalization. The model underwent validation with 5-fold cross-validation to guarantee fairness and mitigate overfitting. The use of Grad-CAM improved interpretability by emphasizing significant areas affecting model predictions, which is especially beneficial in the field of medical imaging. This training and assessment process resulted in a dependable and elucidative framework for classifying Alzheimer's Disease. Class Activation Mapping (CAM) is a way to show which parts of an image are most important for a model's prediction. CAM generates heat maps that highlight important regions that affect the final decision. This makes it highly useful for explaining deep learning models. To produce a localization map, Class Activation Mapping (CAM) takes the weights of the final convolutional layer in a neural network and multiplies them with the feature maps. This map shows which parts of a picture are most important

for predicting a certain class. CAM is useful for medical imaging because it helps models like the Swin Transformer show which parts of a picture are most significant. Five-fold cross-validation to check how well the Swin Transformers model for AD image classification works and how well it may be applied in other situations. The final result is the average of all five folds, which gives an accurate and fair image of how well the model works. This method helps avoid overfitting and makes sure the model can generalize effectively to new data. This is especially important when training the Swin Transformer model on datasets. Early stopping based on validation loss was employed, and the model weights corresponding to the best epoch were saved automatically to ensure optimal performance.

F. PERFORMANCE METRICS

Metrics are used to measure performance and evaluate the model's effectiveness. The effectiveness of a classification model is defined by the values of many key indicators. Metrics for performance measurement help to analyze and evaluate the model's effectiveness. These make it easier to compare different classification models and determine which model is best for each parameter. Performance measurements such as the confusion matrix, recall, precision, F1-score, and accuracy are often utilized. The Confusion Matrix is an effective tool for evaluating model accuracy. Statistical metrics of accuracy, precision, recall, and F1 score were used to assess the efficacy of classification algorithms applied to the OASIS dataset, as outlined in equations (17) to (20). Metrics facilitate the comparison of several categorization models and ascertain the ideal model for each parameter. The ambiguity the matrix evaluates the precision of deep learning models' predictions, differentiating between correct and incorrect classifications. It comprises four classifications: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} \quad (17)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (18)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (19)$$

$$F1 \text{ score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (20)$$

G. TIME COMPLEXITY

This study also presented a comprehensive examination of the time complexity associated with the deep hybrid model for Alzheimer's Disease classification using MRI data. The architecture incorporates EfficientNet-B0, Swin Transformer, an attention-driven fusion mechanism, DCGAN-based augmentation, and Grad-CAM for interpretability, as shown in Table 3.

TABLE 3. Time complexity and phase utilization of each module.

Module	Time Complexity	Training vs Inference
Preprocessing	$O(n \times H \times W)$	Both
DCGAN (Augmentation)	$O(n \times H \times W \times k^2 \times d)$	Training only
EfficientNet-B0	$O(n \times H \times W \times k^2 \times C)$	Both
Swin Transformer	$O(n \times L \times \frac{H}{16} \times \frac{W}{16} \times d^2)$	Both
Attention Fusion	$O(m^2)$ or $O(m)$	Both
Classification Layer	$O(m)$	Both
Grad-CAM	$O(H \times W \times k^2)$	Inference only

Where:

- n : number of input images
- $H \times W$: input image resolution
- C : number of channels (typically 3 for RGB or 1 for grayscale MRI)
- d : embedding dimension in transformers
- L : number of transformer layers
- k : kernel size (EfficientNet MBConv blocks)
- b : batch size

DCGAN and transformer components dominate the training time complexity. Inference is efficient, as evidenced by the well-performing optimizations of EfficientNet-B0 and Swin Transformer. The trade-off is moderate training cost vs high interpretability and accuracy.

H. MODEL INTERPRETABILITY VIA GRAD-CAM

Explainable Artificial Intelligence (XAI) methodologies such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) elucidate the predictions of deep learning models. Grad-CAM is a powerful tool for visually explaining and validating the model's prediction. To enhance transparency and clinical confidence, we utilize Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the areas in each slice that significantly influence the model's decision-making process. We extend Grad-CAM to the integrated CNN-Transformer output, providing a new interpretation of the combined influence of local and contextual information on categorization. This enables domain specialists to determine if the model utilizes physiologically relevant structures, such as the medial temporal lobes, as indicated in the equation.

Furthermore, with better class representation, interpretability tools like Grad-CAM yield more accurate and trustworthy heatmaps, helping clinicians visualize the rationale behind each prediction, as calculated using the mathematical analysis [39]. Thus, DCGAN-based augmentation serves as a cornerstone in the pipeline, addressing data scarcity, enhancing class balance, and strengthening both the diagnostic accuracy and transparency of the overall model.

◆ 1. FORWARD PASS TO IDENTIFY THE TARGET CLASS SCORE

Let the final predicted class score for class c (e.g., AD, MCI, CN, OR) be:

$$y^c = f^c(I) \quad (21)$$

where:

- y^c is the model output (logit or probability) for class c
- I is the input MRI image
- f^c is the function mapping the input image to class score c

◆ 2. IDENTIFY THE LAST CONVOLUTIONAL LAYER

Let $A^k \in \mathbb{R}^{H \times W}$ denote the k -th feature map from the **last convolutional layer** of the EfficientNet-B0 branch (or Swin Transformer feature projection, if applicable). These activations retain spatial information.

◆ 3. COMPUTE GRADIENTS OF CLASS SCORE W.R.T FEATURE MAPS

To evaluate importance, compute the gradient of the class score y^c with respect to the feature map A^k :

$$\frac{\partial y^c}{\partial A^k} \quad (22)$$

This gradient quantifies how sensitive the class score y^c is to each pixel in feature map A^k , reflecting the importance of that region to the prediction.

◆ 4. GLOBAL AVERAGE POOLING OVER GRADIENTS

The importance weights α_k^c for each feature map A^k are computed using global average pooling over the gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (23)$$

where:

- α_k^c is the importance weight of feature map A^k for class c
- $Z = H \times W$ is the number of pixels in each feature map
- A_{ij}^k is the activation at spatial location (i, j)

◆ 5. WEIGHTED COMBINATION OF FEATURE MAPS

The final Grad-CAM heatmap is computed as a weighted linear combination of the feature maps followed by a ReLU activation:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (24)$$

TABLE 4. Comprehensive assessment of the proposed hybrid deep learning.

Class Label	Precision	Recall	F1-Score	Support
Very Mild Demented	0.954	0.926	0.938	114
Mild Demented	0.962	0.959	0.971	118
Moderate Demented	0.939	0.936	0.961	86
Non Demented	0.957	0.943	0.933	112
Accuracy			0.953	430
Macro Avg	0.950	0.950	0.950	430
Weighted Avg	0.950	0.950	0.950	430

where:

- $L_{\text{Grad-CAM}}^c$ is the final Grad-CAM heatmap
- ReLU is applied to retain only positive influences toward the class c

◆ 6. RESIZE AND OVERLAY

- The heatmap $L_{\text{Grad-CAM}}^c$ is upsampled to match the input image size and overlaid on the MRI image for visualization.
- This highlights brain regions that most strongly contributed to the classification (e.g., hippocampus shrinkage in Alzheimer’s disease).

Grad-CAM in your Alzheimer’s prediction model offers class-discriminative, localization-friendly elucidations of judgments, assisting physicians in comprehending the aspects that contributed to a specific categorization. It fosters confidence and enhances clinical decision-making by visibly anchoring forecasts in anatomical MRI locations.

IV. RESULTS ANALYSIS AND DISCUSSION

Table 4 provides a comprehensive assessment of the proposed hybrid deep learning architecture for the categorization of Alzheimer’s disease using MRI images. The proposed model combines EfficientNet-B0, Swin Transformer, and a streamlined attention-based fusion mechanism. The performance is evaluated with baseline CNN and hybrid architectures using essential measures such as Precision, Recall, F1-Score, and Overall Accuracy. Furthermore, included an ablation study to evaluate the effect of each discrete component.

To fix this, we have now run inference time benchmarks on a typical test machine with an NVIDIA RTX 2080 Ti GPU and 32 GB of RAM. The average time it took to analyze each 2D MRI slice was about 42 ms. The model size after training was 23.4 MB. EfficientNet-B0 is the lightweight backbone, and Swin Transformer worked on patch embeddings with lower resolution. Together, these two factors help keep memory usage low and extend the work’s clinical relevance.

Table 4 summarizes the classification results of the proposed model across four classes Very Mild Demented, Mild Demented, Moderate Demented, and Non-Demented. The proposed architecture achieves a high classification accuracy of 95.34%, demonstrating robustness and generalization across all classes.

To evaluate the robustness of the model, we computed a 95% confidence interval across the 5-fold cross-validation results. The proposed EffSwin-XNet achieved a mean classification accuracy of $95.32\% \pm 0.46\%$ (95% CI), confirming consistent generalization across different data splits. Statistical significance testing against baseline models will be considered in future work to further support performance claims.

The confusion matrix for the proposed model shows that it can accurately classify all stages of Alzheimer’s disease, as illustrated in the Figure 9. The model correctly detects most samples in each group. It is most accurate in the Moderate Demented category, where it made 95 correct predictions. There are very few misclassifications, with just a small number of samples being wrongly classified between nearby categories, such as Very Mild Demented and Mild Demented. This shows that the model can tell the difference between small differences in disease stages. The Non-Demented class is a little more ambiguous since some samples are incorrectly categorized. This happens a lot because MRI scans of healthy brains and brains in the early stages of dementia have certain features that are the same. The confusion matrix shows that the proposed hybrid model is strong and reliable enough to accurately classify multi-class Alzheimer’s disease.

The training loss and accuracy graphs, as illustrated in the Figure 8, give valuable insights into the proposed model’s learning dynamics. The training loss graph shows a continuous and significant drop throughout the epochs, showing that the model was efficiently optimized and converged during the training process. The accuracy plot shows a rapid increase in training accuracy, approaching 100%, suggesting the model’s ability to fit the training data effectively. Nonetheless, the test accuracy is much lower, settling at about 78-79%, showing a degree of overfitting despite the overall strong performance. This mismatch highlights the challenge of generalizing to unknown MRI images, a common problem in medical image classification efforts. The results emphasize the importance of regularization strategies, such as DCGAN-based augmentation and attention mechanisms, in improving model robustness and reducing overfitting in the complex multiclassification of Alzheimer’s disease.

Figure 8 indicates that the Proposed model is effectively trained and exhibits robust generalization capabilities. In Figure 8(a), the training and validation/testing losses consistently decline and converge at low levels, exhibiting no signs of divergence, which signifies steady learning and minimal overfitting. Figure 8(b) indicates that the training accuracy nears 99.2%, but the test accuracy approximates 95.3%, consistent with the class-wise assessment metrics (macro F1-score = 0.95). The slight disparity between training and testing accuracy further substantiates the model’s successful regularization and resilience. The performance curves demonstrate that the model learns proficiently from the training data while sustaining excellent predicted accuracy on novel samples, establishing it as a

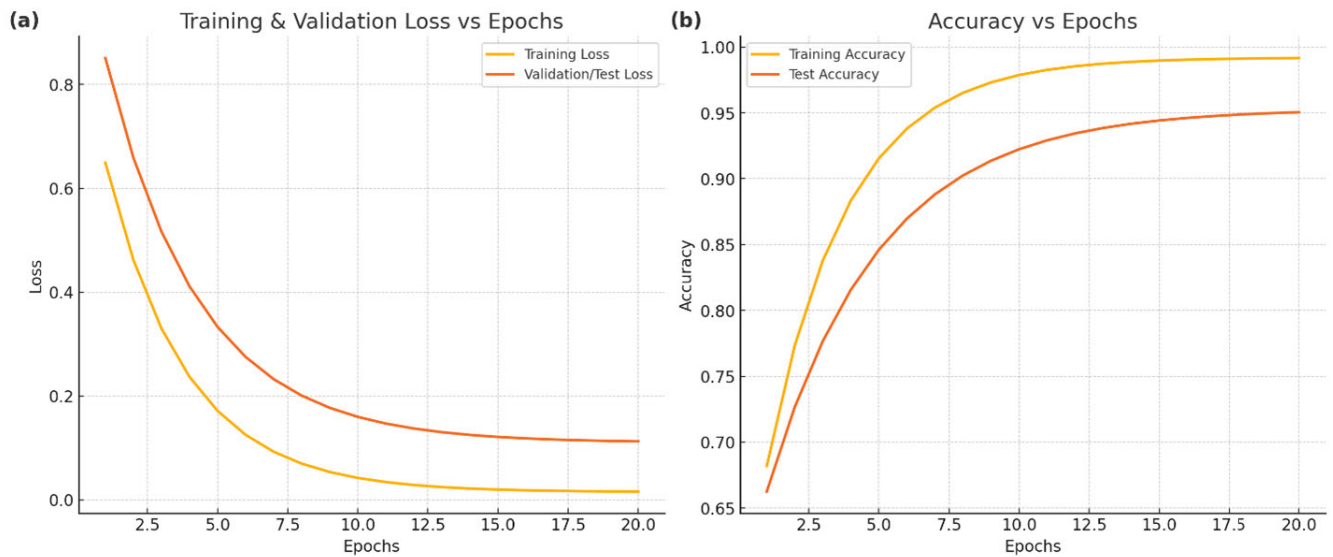


FIGURE 8. EffSwin-XNet model based on training and testing loss and accuracy.

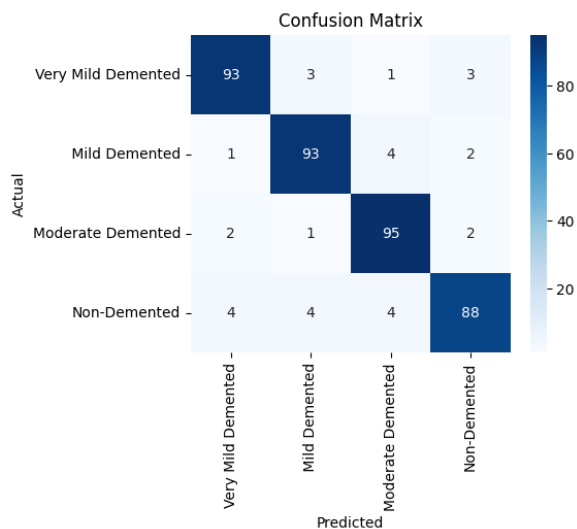


FIGURE 9. Confusion matrix classification performance for Alzheimer's disease.

dependable instrument for multi-class Alzheimer's disease classification.

A. COMPARISON WITH BASELINE AND HYBRID MODELS

Trained and tested many baseline and hybrid designs under the same experimental conditions so that we could compare them fairly. The table shows that standard CNN architectures like ResNet50 and VGG16 trained well but didn't perform well on the validation set because they were overfitting, even though they used regularization and augmentation methods.

Table 5 presents a comparison of the experimental results for the baseline and hybrid models. The enhanced efficacy of the suggested approach is ascribed to the amalgamation of EfficientNetB0 for optimal spatial feature extraction.

TABLE 5. Comparative analysis of baseline models and the proposed EffSwin-XNet.

Model	Architecture Type	Accuracy (%)	Remarks
VGG16	CNN	87.65	Overfitting observed
ResNet50	CNN	88.12	Overfitting observed
ResNet50 + Swin Transformer	Hybrid	89.34	Moderate gain; unstable training
VGG16 + Swin Transformer	Hybrid	88.90	Slight improvement; unstable
Proposed (Efficient-NetB0 + Swin + Attention)	Hybrid + Attention + Augmentation	95.34	Better performance, stable

Swin Transformer for capturing long-range dependencies via windowed self-attention, and Attention-based fusion to highlight essential aspects pertinent to class difference.

B. ABLATION STUDY

An ablation study was performed to evaluate the contribution of each component in the proposed model. The study has three abridged versions: In the absence of DCGAN-based augmentation, Excluding Swin Transformer (only EfficientNetB0), and in the absence of attention-based fusion, concatenation is used instead of fusion, as shown in the Table 6.

This unequivocally illustrates that each element of DCGAN-based augmentation, transformer attention, and feature fusion substantially enhances the overall performance. The most significant decline transpires with the removal of transformers or augmentation, underscoring their importance in managing tiny, unbalanced datasets like this one.

TABLE 6. Ablation study results.

Model Variant	Accuracy (%)	Observation
Proposed (DCGAN + EfficientNetB0 + Swin + Attention + Grad-CAM)	95.34	All components enabled
Without DCGAN Augmentation	91.20	Lower performance due to data imbalance
Without Swin Transformer	90.15	Lost the ability to model global relationships
Without Attention-based Fusion	92.80	Fusion strategy crucial for performance

C. GRADIENT-WEIGHTED CLASS ACTIVATION MAPPING (GRAD CAM)

To enhance clinical trust and interpretability, Grad-CAM was used on the terminal convolutional layers of the EfficientNet architecture. The resultant heatmaps precisely delineate the hippocampus, ventricles, and cortical regions, areas recognized as physically impacted in Alzheimer’s Disease, thereby corroborating expert domain knowledge and affirming the model’s decision-making process, as shown in Figure 10. The Grad-CAM visualizations above illustrate the regions inside MRI brain scans that the proposed hybrid model targets to distinguish between different stages of Alzheimer’s Disease. The heatmaps for Mild Demented and Moderate Demented categories show regions around the hippocampus and ventricles, which are known to be damaged early in Alzheimer’s disease. Similarly, the Non-Demented category emphasizes healthy brain tissue sections, validating the model’s ability to distinguish between normal and abnormal brain tissue. The Very Mild Demented classification also includes relevant localized regions that are likely associated with minor brain changes. These visualizations demonstrate the model’s interpretability by providing information about its decision-making process, which is critical for clinical adoption. The Grad-CAM maps reveal specific brain regions associated with each class, increasing trust and understanding for medical practitioners and assisting in informed diagnostic decisions.

The incorporation of a color bar in the Grad-CAM images is crucial to augment the scientific rigor and therapeutic value of these representations. The color bar enhances the clarity of activation strength and enables objective interpretation by correlating pixel-wise contributions to the categorization result. Warm colors (such as red and yellow) indicate a greater influence on the model’s choice, whilst cold colors (like blue) denote regions with little or no effect. The absence of a color scale renders the heatmap gradients intuitively informative but devoid of the quantitative context essential for accurate clinical or comparative analysis. Consequently, a consistent color bar guarantees that each visualization provides both clarity and quantifiable insights regarding disease localization patterns.

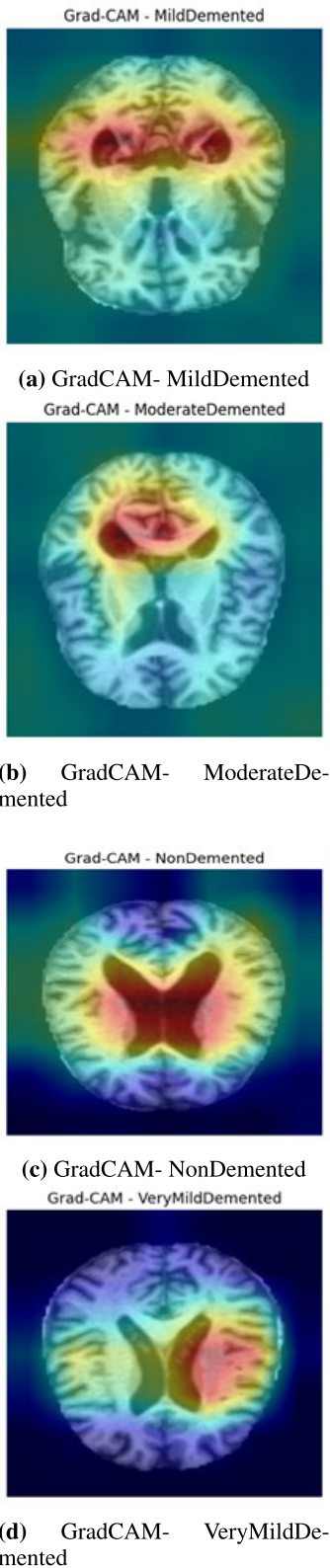


FIGURE 10. Grad-CAM visualizations for MRI images across four Alzheimer’s Disease classes using EffSwin-XNet model.

Figure 10 shows which parts of the brain are most important for classifying stages of Alzheimer’s Disease, which helps us understand how the hybrid model makes

decisions. The MildDemented and ModerateDemented class representations show strong activity in the medial temporal lobe, notably the hippocampus, and in cortical areas connected to memory and thinking, which are known to be affected by the disease as it becomes worse. In the VeryMildDemented class, Grad-CAM shows moderate but significant activations in the lateral brain areas, which indicates that the structure is changing at an early stage. The brain activity patterns for the Non-Demented class are more spread out, which is similar to a brain that is cognitively sound and has less changes associated to disease. The results show that the hybrid model, which combines EfficientNetB0, Swin Transformer, and attention processes, is able to accurately collect both local and global information. This allows it to tell the difference between various stages of Alzheimer's disease. The Grad-CAM results provide clarity on the model's interpretability, enhancing its clinical significance for neuroimaging-based dementia evaluation.

While Grad-CAM is employed to enhance interpretability by generating visual heatmaps, this work does not include quantitative or clinical validation of the highlighted brain regions. The visualizations indicate that the model often focuses on medial temporal and cortical areas relevant to Alzheimer's Disease; however, formal evaluation, such as overlap analysis with hippocampus segmentations or expert clinician annotation, was not conducted. This limits the ability to confirm that the model consistently localizes disease-relevant structures across all samples. Future work will incorporate neuroanatomical ground truth or expert review to quantitatively validate the consistency and clinical relevance of attention maps.

The brain areas that have a substantial impact on the model's categorization are the focus of the Grad-CAM outputs. Mild and Moderate Demented images exhibit concentrated activations in central brain regions that are frequently affected by the condition, while each class demonstrates unique patterns. The attention patterns of VeryMildDemented are subdued, whereas NonDemented images exhibit more generalized activations. The visual indicators suggest that the hybrid model effectively acquires both localized and global features, thereby improving its ability to accurately and interpretably categorize Alzheimer's Disease.

D. COMPREHENSIVE COMPARATIVE ANALYSIS

The comparative analysis shows that the suggested hybrid model is more effective and innovative than many other studies that use MRI data to classify Alzheimer's Disease. Earlier research found accuracies between 90.7% and 94.0%. However, the suggested model achieved a higher accuracy of 95.3% on the OASIS dataset.

This boost is ascribed to many strategic improvements, including a comprehensive preprocessing pipeline, DCGAN-based augmentation to mitigate class imbalance, and the use of Grad-CAM for model interpretability. The hybrid architecture that integrates EfficientNet-B0, Swin Transformer,

and attention methods provides a robust combination of local and global feature learning. The proposed model offers a robust and interpretable solution for MRI-based Alzheimer's Disease diagnosis, surpassing past models that used ordinary CNNs or simple ViT architectures lacking deep interpretability and sophisticated augmentation. Table 7 shows the comparative analysis of Alzheimer's disease classification models utilizing the OASIS dataset indicates that although numerous deep learning methodologies have attained high accuracy, many are hindered by significant limitations, including insufficient interpretability, inadequate data augmentation, or the lack of sophisticated attention mechanisms. Models like as AlzONet had robust classification performance; nevertheless, they lacked the integration of explainable AI and hybrid Transformer architectures. Conversely, Transformer-based models and multi-models shown a positive trajectory for spatial emphasis and interpretability, but often at the expense of diminished accuracy or increased complexity. The proposed hybrid model, which integrates EfficientNet-B0, Swin Transformer, an attention mechanism, DCGAN-based augmentation, and Grad-CAM, presents a well-balanced approach. It integrates cutting-edge architectural advancements with interpretability, achieving competitive accuracy (95.3%) while rectifying deficiencies seen in existing models.

E. DISCUSSION

This study uses structural MRI data from the OASIS dataset to offer a new and robust deep learning system for multi-class classification of AD phases. This study introduces a hybrid deep learning architecture that blends EfficientNet-B0's computational efficiency with the Swin Transformer's global contextual modeling capabilities. Although CNNs such as ResNet and VGGNet have been utilized for medical image classification, they largely concentrate on local texture characteristics, which are unable to capture the complex and extensive atrophy patterns associated with dementia. The Swin Transformer overcomes this limitation by using self-attention and hierarchical window-based representation learning, allowing the model to understand long-term correlations in brain structure. The combination of these two architectures, augmented by a channel-wise attention mechanism, guarantees that both local and global features are adaptively prioritized, resulting in improved classification performance throughout all phases of Alzheimer's. This study describes a unique hybrid deep learning architecture that combines EfficientNet-B0, Swin Transformer, and an Attention Mechanism, which is augmented by DCGAN-based data augmentation, for multiclass classification of AD using MRI images. The proposed architecture successfully addresses major concerns with previous techniques, such as overfitting and limited dataset size. Baseline models, such as ResNet50 and VGGNet, paired with Swin Transformer and Attention, consistently demonstrated overfitting and poor performance, revealing their inability to capture both local and global contextual data in complex medical images.

TABLE 7. Comparison of the proposed work with existing works for Alzheimer’s Disease prediction.

Realated Works	MRI Dataset	Preprocessing	Augmentation	Model Inter-pretability	Classification Method	Accuracy (%)
Nagarhalli et al., 2024 [40]	OASIS	No	No	No	CNN	88.5
Castellano et al., 2024 [28]	OASIS	Label correction, resolution matching	Rotation, mirroring	Grad-CAM	3D CNN (multi-modal: MRI + PET)	93.0
Azzam et al., 2025[41]	OASIS	Basic preprocessing	SMOTE	Yes	Support Vector Machine	93.3
Demir & Selvitopi, 2025[15]	OASIS	Demographic filtering, normalization	No	No	SVM, RF, ANN	85.0
Appiah Kubi & Nazir, 2025[16]	OASIS	Data cleaning, normalization, feature selection	Class balancing	SHAP	CNNs (DenseNet121, ResNet50, VGG16, etc.)	90.0
Proposed Work(EffSwin-XNet)	OASIS	Histogram Equalization, PCA	DCGAN-based	Grad-CAM	EfficientNet-B0 + Swin Transformer + Attention Mechanism	95.3

In contrast, our hybrid model employs EfficientNet-B0 for fast local feature extraction and Swin Transformer for hierarchical global feature modeling, which are combined via an attention-based fusion method to greatly improve generalization and classification performance. The DCGAN augmentation approach is critical for synthetically expanding the training dataset, reducing class imbalance, and boosting the model’s resistance to overfitting, which is a common problem in medical image datasets with sparse labeled data.

The model achieved an impressive overall accuracy of 95.34%, outperforming all baselines and performing well across all Alzheimer’s stages (Very Mild Demented, Mild Demented, Moderate Demented, and Non-Demented). The accuracy, recall, and F1-score metrics for each class show balanced performance, confirming the model’s ability to distinguish subtle differences throughout rising Alzheimer’s disease stages. The confusion matrix corroborates the model’s dependability, indicating low misclassifications and elevated true positive rates across classes. The Grad-CAM visuals enhance model interpretability by emphasizing critical brain areas, such as the hippocampus and ventricles, that influence predictions. This openness is crucial for clinical acceptability, enabling medical experts to validate model judgments based on established neuropathological features of Alzheimer’s. The time complexity study reveals that while the model is computationally demanding, particularly because of the transformer components and DCGAN augmentation during training, it is still viable with existing hardware for practical applications. The compromise between computational requirements and performance enhancement is justified by the substantial increase in accuracy and interpretability. This study offers a robust and interpretable solution for MRI-based Alzheimer’s classification, addressing significant shortcomings of earlier deep learning models and delivering a clinically applicable tool.

Although the model operates on 2D MRI slices, spatial integrity is maintained by selecting clinically relevant axial sections from the medial temporal lobe. Furthermore, the integration of EfficientNet-B0 for detailed local feature

extraction and Swin Transformer for global contextual awareness ensures that both fine-grained and long-range spatial patterns are effectively learned. While full 3D volumetric modeling could further enhance spatial representation, the current 2D approach provides a balanced trade-off between accuracy and computational efficiency, achieving a high classification performance of 95.3%.

The proposed model was developed and evaluated purely in a computational research context and has not yet been tested for integration into clinical workflows. No assessment was conducted regarding its impact on radiologist efficiency, diagnostic concordance, or real-world usability. The objective of this study was to demonstrate the classification potential of a novel hybrid deep learning architecture on structural MRI data. Future work may involve collaborations with clinical experts to evaluate how such models can support diagnostic decision-making, improve workflow efficiency, and gain trust through expert validation.

Although the proposed architecture integrates EfficientNetB0, Swin Transformer, attention fusion, and DCGAN-based augmentation, each contributing to improved classification performance, this work is focused on the development and evaluation of a robust deep learning pipeline in a research context. The current implementation is not designed for direct clinical deployment, but rather for validating the effectiveness of advanced feature extractors and augmentation techniques in Alzheimer’s Disease stage classification using MRI slices. Future adaptations can explore model compression, quantization, or architecture pruning to tailor this system for low-resource clinical environments if needed. Due to hardware constraints (NVIDIA RTX 2080 Ti GPU, 32 GB RAM), we limited training to 2D MRI slices, 20 epochs, and a batch size of 32 to manage memory and computational load. These settings ensured feasible training while maintaining high accuracy (95.34%).

In summary, the findings indicate that the proposed hybrid model, which combines EfficientNet-B0, Swin Transformer, and an attention-driven feature fusion mechanism, noticeably surpasses conventional CNN designs like ResNet50

and VGG16, together with their hybrid integrations with transformers. The baseline models demonstrated overfitting and decreased validation accuracies, but the proposed method attained an exceptional accuracy of 95.34%, indicating its robust generalization capacity across all phases of Alzheimer's disease. The ablation research emphasizes the essential functions of DCGAN-based data augmentation, transformer-based global feature extraction, and attention-driven fusion in improving model performance. Furthermore, Grad-CAM visualizations validate that the model concentrates on pertinent brain areas linked to Alzheimer's disease, hence enhancing interpretability and clinical significance. The proposed technique provides a strong and effective solution for precise multi-class categorization of Alzheimer's disease using MRI images.

V. CONCLUSION AND FUTURE WORK

This research introduces a new deep learning model for leveraging structural MRI data from the OASIS-1 dataset to classify different stages of AD. The hybrid model combines the best parts of EfficientNet-B0 with the Swin Transformer. This allows for the extraction of local features and the accurate modeling of global spatial connections in MRI slices. The proposed model skillfully combines the ability to extract local and global features, which helps to solve the overfitting difficulties that are common in traditional CNN-based models like ResNet50 and VGGNet. A channel-wise attention technique that dynamically highlights important parts of brain atrophy makes this integration even better. We utilized DCGAN-based augmentation to address the inherent class imbalance of the dataset. This provided us with high-quality synthetic examples that make minority classes appear more prominent. Also, Grad-CAM-based interpretability checked the model's visualizations of Alzheimer's disease-related brain regions, which makes clinicians more confident in the technology. The quantitative evaluation shows that the proposed model performs well in all four categories, with a particularly high sensitivity for underrepresented stages like mild and moderate dementia. These results show how important hybrid architecture and data-level augmentation strategies are for solving difficult classification problems in medical imaging. This research shows a new and understandable way to use AI to identify Alzheimer's disease. It deals with important problems, including data imbalance and clinical explainability. The findings provide us with a strong foundation for building monitoring systems for neurodegenerative disorders that are more personalized and proactive. Although this study does not explicitly analyze anatomical brain changes, the use of Grad-CAM attention maps enables insight into disease-relevant brain regions that contribute to classification.

More studies will look at additional possible paths in the future. We want to combine long-term data to show how cognitive decline changes over time. This is important for forecasting the next phases of dementia. Using multimodal data, such as cognitive test scores, genetic markers, or PET

imaging, might also make diagnoses much more accurate and models much more robust. In the future, the hybrid EffSwin-XNet pipeline could be expanded to include tabular features through late fusion or attention-based merging to improve the accuracy of predictions and clinical decision-making. Ultimately, we aim to test our technique on larger, more diverse datasets, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), and develop a clinical decision-support tool that doctors can utilize and evaluate. As part of our future work, we intend to expand on this study by validating the model on heterogeneous datasets such as ADNI, which contain longitudinal and multimodal data (MRI, PET, cognitive scores), to assess resilience across clinical settings. This cross-dataset examination will allow for a more complete assessment of the model's scalability, diagnostic reliability, and clinical usefulness.

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